Solution 1 Swapping Φ^+ and Φ^-

Problem 1.1 — Prove that there is a unique element σ in the Weyl group W_Φ sending Φ^+ to Φ^-.

Recall that W_Φ acts simply transitively on the set of all bases of Φ^+ by the theorem on page 51 of Humphreys. Now, if Δ is a basis, then $\{-v : v \in \Delta\}$ is a basis (this is obvious) thus there exists a unique $\sigma \in W_\Phi$ such that $\sigma(\Delta) = \{-v : v \in \Delta\}$.

Problem 1.2 — Prove that any reduced expression for σ must involve all σ_α, $\alpha \in \Delta$.

Suppose σ_α is never used for some $\alpha \in \Delta$. By lemma 10.2B, σ_β merely permutes the elements of $\Phi^* \setminus \{\beta\}$ for all $\beta \in \Delta$. Thus, $\sigma = \prod \sigma_{\beta_i}$ will never send α into Φ^-.

Solution 2 Dual Root System

Problem 2.1 — Let $\Phi^\vee$ be the dual root system of Φ and $\Delta^\vee = \{\alpha^\vee | \alpha \in \Delta\}$. Prove that $\Delta^\vee$ is a base of $\Phi^\vee$.

This is really silly. The mapping from $\alpha \to \alpha^\vee = \frac{2\alpha}{(\alpha|\alpha)}$ simply scales each root by a constant, positive factor. This doesn't change the fact that all elements of $\Phi_+^\vee$ (which is the same positive half-space corresponding to γ; everything is just scaled!) can be written as positive (real) linear combinations of elements of $\Delta^\vee$.

Okay, but then $\Delta^\vee$ has to be the (unique, existent) basis of $\Phi_+^\vee$, because no other subset of $\Phi_+^\vee$ could play this role. Indeed, if you picked a $\Delta_0^\vee \neq \Delta^\vee$, such that $\alpha \in \Delta^\vee \setminus \Delta_0^\vee$, then observe than when written in the $\Delta^\vee$ basis, any positive linear combination of the elements of $\Delta_0^\vee$ which has a positive α-component must have another nonzero component.

If you want me to write down some symbols, here you go:

$$\begin{aligned}
\beta^\vee &= \frac{2}{(\beta|\beta)}\beta \\
&= \frac{2}{(\beta|\beta)} \sum_{\alpha \in \Delta} c_\alpha \alpha \\
&= \sum_{\alpha \in \Delta} \frac{2c_\alpha}{(\beta|\beta)} \alpha \\
&= \sum_{\alpha \in \Delta} \frac{2c_\alpha}{(\beta|\beta)} \frac{2\alpha^\vee}{(\alpha^\vee|\alpha^\vee)} \\
&= \sum_{\alpha \in \Delta} \frac{c_\alpha(\alpha|\alpha)}{(\beta|\beta)} \alpha^\vee
\end{aligned}$$

where c_α are positive integers.

Then, write the elements of $\Phi_+^\vee$ in the basis $\Delta^\vee$. Elements are either:

- Exactly α.
- Do not have a positive α-component.
- Have a positive α-component, but also have another positive component.

How in the world are you supposed to write α as a positive linear combination of elements of the second two types?

Solution 3 Irreducible Root Systems

Problem 3.1 — Let Φ be irreducible. Prove that $\Phi^\vee$ is also irreducible.

How hard is it to understand that $\Phi^\vee$ is just Φ with all vectors scaled by positive factors? Clearly if nothing was orthogonal to anything else before, nothing will be orthogonal to anything else now!

$$(\alpha^\vee, \beta^\vee) = \frac{4}{(\alpha|\alpha)(\beta|\beta)}(\alpha, \beta) \neq 0$$

Problem 3.2 — If Φ has all roots of equal length, prove that $\Phi^\vee$ is isomorphic to Φ.

Everything is scaled by the *same* amount, $c = \frac{2}{(\alpha|\alpha)}$. This is clearly an isomorphism; $\langle\beta|\alpha\rangle$ is dimensionless.

$$\langle\sigma(\alpha)|\sigma(\beta)\rangle = \frac{2c^2(\alpha|\beta)}{c^2(\beta|\beta)} = \langle\alpha|\beta\rangle$$

Problem 3.3 — Suppose Φ has two root lengths. Show that if $\alpha \in \Phi$ is long, then $\alpha^\vee$ is short and vice versa.

Are these questions even real. Suppose your lengths were originally $\|\alpha\| = a < \|\beta\| = b$. Then, $\|\alpha^\vee\| = \frac{2a}{(\alpha|\alpha)} = \frac{2}{a} > \frac{2}{b} = \|\beta^\vee\|$.

Problem 3.4 — Suppose Φ has two root lengths. Show that Φ has a unique maximal short root.

wut we waffling abt rn. I'll show $\Phi^\vee$ has a unique maximal short root, and the problem follows by flipping $\Phi^\vee$ back to Φ.

Recall from problem 2 that we can write all $\beta^\vee \in \Phi_+^\vee$ as

$$\beta^\vee = \sum_{\alpha\in\Delta} \frac{c_\alpha(\alpha|\alpha)}{(\beta|\beta)}\alpha^\vee$$

where c_α and $d_\alpha = \frac{c_\alpha(\alpha|\alpha)}{(\beta|\beta)}$ are positive integers.

Now, I know that all lengths are either a or $b > a$. Recall that Φ_+ automatically admits a unique maximal root β, which must be long. This means $\beta - \alpha$ is a positive linear combination of Δ for all $\alpha \in \Phi_+$. After conjugation, the claim is that $\beta^\vee - \alpha^\vee$ is a positive integer linear combination of $\Delta^\vee$ for all short $\alpha^\vee$. Well, I already know that $\beta^\vee - \alpha^\vee = \frac{2}{a^2}(\beta - \alpha)$ is a *positive* linear combination of $\Delta^\vee$ for all short $\alpha^\vee$, so the claim is that this is an *integer* combination, but this is true by default because $\Delta^\vee$ is a basis.

Solution 4 Dual Root Systems of Irreducible Root Systems

Problem 4.1 — Prove that each irreducible root system is isomorphic to its dual root system, except for B_ℓ and C_ℓ.

Show that the root systems of B_ℓ and C_ℓ are dual to each other.

Recall that root systems are uniquely determined up to isomorphism by their Dynkin diagrams. Recall also that irreducible root systems come in two forms:

- If all roots have the same length, then they are clearly isomorphic to their dual; all vectors are scaled by the same amount. This occurs in root systems of type A_ℓ, D_ℓ, E_6, E_7, and E_8.
- If there are two root lengths, then the dual root system sends $v \to \frac{2v}{(v|v)}$. In other words, long roots become short roots, and vice versa. Now, observe that

  $$\langle a^\vee, b^\vee \rangle \langle b^\vee | a^\vee \rangle = \langle a|b \rangle \langle b|a \rangle$$

  by direct calculation. Thus, the Dynkin diagram of the dual root system is the same as the Dynkin diagram of the original root system, but with all arrows reversed. This swaps root systems of type B_ℓ and C_ℓ, and preserves root systems of type F_4 and G_2.

Solution 5 Automorphisms of Root Systems

Problem 5.1 — Prove that the map $\alpha \mapsto -\alpha$ is an automorphism of Φ. Decide for which irreducible Φ this automorphism of Φ belongs to the Weyl group.

The map is an automorphism because $(-a, -b) = (a, b)$.

Now, figuring out whether or not this is in the Weyl group is a matter of massive amounts of bashing. First off, by the table on page 66 of Humphreys, we know that the Weyl group is in fact equal to the automorphism group for all root systems other than A_ℓ, D_ℓ, and E_6, so clearly the total reflection belongs to the Weyl group.

Now, consider A_ℓ. The standard basis for A_ℓ embeds this group in the hyperplane $E = \{v_1 + \dots v_{\ell+1} = 0\} \subset \mathbb{R}^{\ell+1}$, written

$$\Delta = \{\underbrace{e_1 - e_2}_{\alpha_1}, \dots, \underbrace{e_\ell - e_{\ell+1}}_{\alpha_\ell}\}$$

Then, the reflection with respect to α_i simply permutes subscripts i and $i+1$, and these transpositions are supposed to generate the Weyl group W, thus W is merely all permutations of the subscripts $1, \dots, \ell+1$.

Now, there are two automorphisms of $\mathbb{R}^{\ell+1}$ which invert E. One is $-I_{\ell+1}$, which is clearly not a permutation. The other one preserves $(1, \dots, 1)^\top$ while flipping E. This is the matrix

$$\begin{pmatrix} \frac{2}{\ell+1} - 1 & \frac{2}{\ell+1} & \cdots & \frac{2}{\ell+1} \\ \frac{2}{\ell+1} & \frac{2}{\ell+1} - 1 & \cdots & \frac{2}{\ell+1} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{2}{\ell+1} & \frac{2}{\ell+1} & \cdots & \frac{2}{\ell+1} - 1 \end{pmatrix}$$

This is not a permutation matrix unless $\ell + 1 = 2$, or $\ell = 1$. Thus, the total reflection is an element of the Weyl group of A_1, but not of A_ℓ for $\ell > 1$.

Next, consider D_ℓ. The standard basis for D_ℓ lives in $E = \mathbb{R}^\ell$, with standard basis

$$\Delta = \{\underbrace{e_1 - e_2}_{\alpha_1}, \dots, \underbrace{e_{\ell-1} - e_\ell}_{\alpha_{\ell-1}}, \underbrace{e_{\ell-1} + e_\ell}_{\alpha_\ell}\}$$

The first $\ell - 1$ basis vectors create all permutations of the subscripts $1, \dots, \ell$ as before. By using the last element, we can cause any combination of an *even number of sign flips* (this is record in Humphreys on page 65).

Well, we are trying to create the automorphism $-I_\ell$. This is possible if and only if ℓ is even. Thus, the total reflection is an element of the Weyl group of D_ℓ if and only if ℓ is even.

Finally, consider E_6. I have no idea how to solve this problem, so I wrote the following code:

```
"""
Today, we will be explicitly enumerating the elements of the Weyl group of E6.

We will work in the following embedding inside of R^9:

One possible selection for the simple roots of E6 is:

(0,0,0;0,0,0;0,1,-1)
(0,0,0;0,0,0;1,-1,0)
(0,0,0;0,1,-1;0,0,0)
```

```
(0,0,0;1,-1,0;0,0,0)
(0,1,-1;0,0,0;0,0,0)
(1/3, -2/3, 1/3; -2/3, 1/3, 1/3, -2/3, 1/3, 1/3)

The Weyl group of E6 is generated by reflections in the
hyperplanes orthogonal to these roots.
"""

import numpy as np

SIMPLE = [
    np.array([0, 0, 0, 0, 0, 0, 0, 1, -1]),
    np.array([0, 0, 0, 0, 0, 0, 1, -1, 0]),
    np.array([0, 0, 0, 0, 1, -1, 0, 0, 0]),
    np.array([0, 0, 0, 1, -1, 0, 0, 0, 0]),
    np.array([0, 1, -1, 0, 0, 0, 0, 0, 0]),
    np.array([1/3, -2/3, 1/3, -2/3, 1/3, 1/3, -2/3, 1/3, 1/3])
]


SIMPLE_REFLECTIONS = [np.identity(
    9) - 2 * np.outer(s, s) / np.dot(s, s) for s in SIMPLE]

WEYL_GROUP = [np.identity(9)]
WEYL_EXPANSION = [()]

DISC = 2 ** 16

WEYL_GROUP_SET = set([tuple(int(i) for i in w.reshape(81) * DISC + 0.5)
                      for w in WEYL_GROUP]) # for fast membership testing

i = 0
while i < len(WEYL_GROUP):
    for j, s in enumerate(SIMPLE_REFLECTIONS):
        test = np.matmul(s, WEYL_GROUP[i])
        test_tuple = tuple(int(i) for i in test.reshape(81) * DISC + 0.5)
        if test_tuple not in WEYL_GROUP_SET:
            WEYL_GROUP.append(test)
            WEYL_GROUP_SET.add(test_tuple)
            WEYL_EXPANSION.append(WEYL_EXPANSION[i] + (j,))
    i += 1

print("The Weyl group of E6 has cardinality " + str(len(WEYL_GROUP)) + ".")

# Find the unique element of the Weyl group which sends
# The simple roots to some permutation of the inverses of the simple roots.

res = None

for n, w in enumerate(WEYL_GROUP):
    NEW_SIMPLE = [np.matmul(w, s) for s in SIMPLE]
    found_all = True
    for i in NEW_SIMPLE:
        found = False
        for j in SIMPLE:
            if np.allclose(i, -j):
                found = True
```

```
                break
        if not found:
            found_all = False
            break
    if found_all:
        res = n
        break

print("The unique element of the Weyl group which " +
      "sends the simple roots to their inverses is:")
print("1/3 * \n" + np.round(3 * WEYL_GROUP[res], 3).__str__())
print("This element can be expanded as:")
print(WEYL_EXPANSION[res])
# print("As matrices:")
# for i in WEYL_EXPANSION[res]:
#     print(SIMPLE_REFLECTIONS[i])

print("This permutes the roots as follows:")

permutation = []
for i in SIMPLE:
    for j, k in enumerate(SIMPLE):
        if np.allclose(i, -np.matmul(WEYL_GROUP[res], k)):
            permutation.append(j)
            break

print(permutation)
```

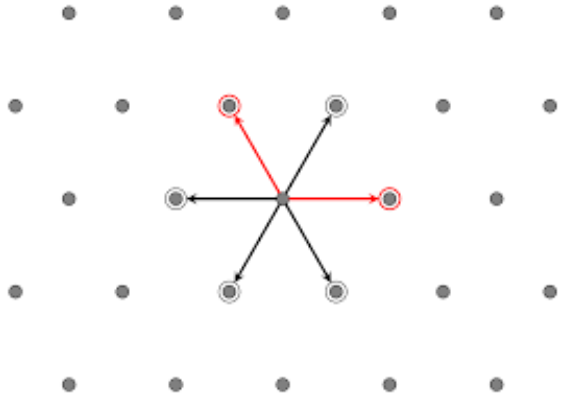
It shows that the total reflection is not an element of the Weyl group of E_6. It also outputs the unique string of reflections that sends the set of positive simple roots to the set of negative simple roots, which is *not* the total reflection.

> **Problem 5.2 —** Prove that an inclusion of one Dynkin diagram in another induces an inclusion of the corresponding root systems.

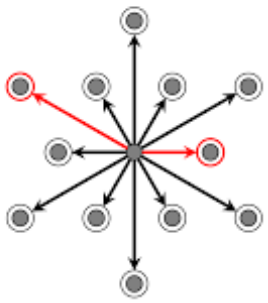
Well, the nodes in Dynkin diagrams represent basis roots, so given a diagram inclusion, we just take the big basis and stick the smaller basis into the larger basis correspondingly. What more is there to say? All the dot products are preserved, since this information is literally contained inside the Dynkin diagram.

> **Problem 5.3 —** Prove that the long roots in G_2 form a root system in E of type A_2.

Well, the long roots in G_2 are form a hexagon, which is exactly A_2.



G_2 ( ) root system:



Solution 6 Root Systems of $\mathfrak{so}_{2\ell+1}$

Problem 6.1 — Prove that B_ℓ, $\ell \geq 2$ is isomorphic to the root system of $\mathfrak{so}_{2\ell+1}$.

Let us recall that $\mathfrak{so}_{2n+1}$ can be represented by skew-symmetric $(2n+1) \times (2n+1)$ matrices. Specifically, we can choose a basis for $\mathfrak{so}_{2n+1}$ consisting of the elements

$$M_{ij} \equiv E_{ij} - E_{ji}, \quad 1 \leq i < j \leq 2n+1$$

where E_{ij} denotes the matrix with a 1 in the (i,j) entry and 0 elsewhere.

Claim (Diagonalizing $\mathrm{ad}_{M_{ij}}$) — The $\binom{2n+1}{2}$ eigenvectors of $\mathrm{ad}_{M_{ij}}$ are given by the following matrices (taking $i = 1, j = 2$ and $n = 2$ for demonstration purposes):

$$\begin{pmatrix} 0 & & 1 & & \\ & 0 & \pm i & & \\ -1 & \mp i & 0 & & \\ & & & 0 & \\ & & & & 0 \end{pmatrix}, \begin{pmatrix} 0 & & 1 & & \\ & 0 & \pm i & & \\ & & 0 & & \\ -1 & \mp i & & 0 & \\ & & & & 0 \end{pmatrix}, \begin{pmatrix} 0 & & & & 1 \\ & 0 & & & \pm i \\ & & 0 & & \\ & & & 0 & \\ -1 & \mp i & & & 0 \end{pmatrix}$$

(there are $2(2n-1)$ of these), in addition to the $\binom{2n-1}{2} + 1$-dimensional kernel

$$\begin{pmatrix} & c & \\ -c & & \\ & & \mathfrak{so}(2n-1) \end{pmatrix}$$

Note that we have listed $2(2n-1) + \binom{2n-1}{2} + 1 = \binom{2n+1}{2}$ eigenvectors, so these indeed form an eigenbasis.

Proof: This can be verified by inspection or by multiplying out the matrices. It can be observed that everything in the first list has eigenvalue $\pm i$, while everything in the second list has eigenvalue 0. □

Claim (Definition of the Cartan subalgebra) — The set of matrices

$$H = \{M_{12}, M_{34}, \cdots, M_{2n-1,2n}\}$$

forms a basis for the Cartan subalgebra of $\mathfrak{so}_{2n+1}$.

Proof: It can be observed that the commutator of any matrix M_{ij} with another matrix M_{kl}, where i, j, k, l are distinct, results in a matrix of the form

$$\begin{pmatrix} & c & \\ -c & & \\ & & \mathfrak{so}(2n-1) \end{pmatrix}$$

This can be verified by explicitly calculating the commutation of M_{ij} with M_{kl} Therefore, it follows that the matrices $M_{12}, M_{34}, \cdots, M_{2n-1,2n}$ commute with each other. Furthermore, their joint kernel is is precisely the span of $M_{12}, \ldots M_{2n-1,2n}$ indicating that no additional elements can be added to the subalgebra. We already showed that all of these matrices are diagonalizable, so they form a basis for the Cartan subalgebra. □

Now we need to determine the roots of this Cartan subalgebra, which means we have to find their common eigenvectors and determine their corresponding eigenvalues. According to the second claim, consider matrices of the following form (for illustrative purposes, let's assume $n = 2$):

$$\begin{pmatrix} 0 & & 1 & \pm_2 i & \\ & 0 & \pm_1 i & \pm_1 \mp_2 1 & \\ -1 & \mp_1 i & 0 & & \\ \mp_2 i & \pm_1 \pm_2 1 & & 0 & \\ & & & & 0 \end{pmatrix}$$

where we have four choices of $(\pm_1, \pm_2) \in \{1, -1\}^2$. The above matrix is an eigenvector of $\mathrm{ad}_{M_{12}}$ with eigenvalue $\pm_1 i$, and an eigenvector of $\mathrm{ad}_{M_{34}}$ with eigenvalue $\pm_2 i$. In general, when we consider different choices of $(\pm_1, \pm_2)$, the above matrix will have eigenvalues of $\pm_1 i$ and $\pm_2 i$ for two of the Cartan subalgebra basis elements, and will be in the kernel of all the other Cartan subalgebra basis elements. Therefore, we obtain roots of the form (expressing covectors in terms of the co-basis corresponding to our Cartan basis)

$$\underbrace{(0, \cdots, 0, \pm_1 i, 0, \cdots, \pm_2 i, 0, \cdots, 0)}_{\pm i\text{'s in any two positions}} \tag{$\star$}$$

There are $4\binom{n}{2}$ of these. Additionally, we have the eigenvectors of the form (again, taking $n = 2$ for demonstration purposes)

$$\begin{pmatrix} 0 & & & & 1 \\ & 0 & & & \pm i \\ & & 0 & & \\ & & & 0 & \\ -1 & \mp i & & & 0 \end{pmatrix}$$

These eigenvectors have eigenvalue $\pm i$ with respect to $\mathrm{ad}_{M_{ij}}$ in the above example, and they are in the kernel of the remaining Cartan subalgebra elements. Therefore, we obtain roots of the form

$$\underbrace{(0, \cdots, 0, \pm i, 0, \cdots, 0)}_{\pm i \text{ in any one position}} \tag{$\star\star$}$$

There are $2n$ of these. In addition to the $4\binom{n}{2}$ roots mentioned earlier, these roots complete the set of roots for the Cartan subalgebra. This can be verified by checking the dimensions: $2n + 4\binom{n}{2} + n = \binom{2n+1}{2}$, which is the dimension of $\mathfrak{so}_{2n+1}$ (including the dimension of the Cartan subalgebra itself).

To complete the problem, we transfer the Killing form from $\mathfrak{so}_{2n+1}$ to the Cartan subalgebra. It is well-known that the Killing form of $\mathfrak{so}_{2n+1}$ in our basis is $\kappa(X, Y) = (2n-1)\operatorname{Tr}(XY)$ (e.g. repeat the proof of problem 4 on pset 4), i.e. $\kappa(M_{ij}, M_{kl}) = (2n-1)(\delta_{jk}\delta_{il} - \delta_{ik}\delta_{jl})$. The restriction to our Cartan subalgebra is just $-(2n-1)$ identically. The constant factor doesn't matter; what matters is that this is the standard fully **negative** definite inner product on $\mathbb{C}^n$.

Transferring this to the dual space, we see that this corresponds to the fully **positive** definite inner product as long as we rotate each root by i. Under this basis, our roots can be written:

- $\pm e_i \pm e_j$ for $i \neq j$ (roots of type ($\star$)).
- $\pm e_i$ for $i \neq j$ (roots of type ($\star\star$)).

This is precisely the root system of type B_n.